Name and Section: Points: / 48

MTH:3340 Abstract Algebra Quiz 3
Due: Friday Feb. 13

Complete all of the following problems being as detailed as you can be. You may work in groups, but you must submit your own solutions using your own words. You must submit a physical copy of your solutions to me unless told otherwise. If there is a specific number of problems which seem unsolvable let me know first, I may have made a typo. Even if you cannot completely solve a problem, a good description of key terms and concepts relevant to the problem is sufficient for some partial credit.

Handwriting: Your ability to express your solutions is just as important as mathematical accuracy. You are expected to write every solution using complete sentences when possible. You will be penalized up to 4 points for notation and disorganized or hard to read solutions.

1. (12 points) Which of the following operations defined on the set $G = \{1, 2, 3, 4\}$ form a group? Explain why or why not in each case.

(a)

$\circ$	1	2	3	4
1	1	2	3	4
2	2	1	4	3
3	3	4	1	2
4	4	3	2	1

$\circ$ is commutative $x \circ y = y \circ x$

1 is the identity.
All elements have an inverse.

$(2 \circ 3) \circ 4 = (3 \circ 2) \circ 4 = 4 \circ (3 \circ 2) = 4 \circ (2 \circ 3) = 4 \circ 4 = 1$
$(2 \circ 4) \circ 3 = (4 \circ 2) \circ 3 = 3 \circ (4 \circ 2) = 3 \circ (2 \circ 4) = 3 \circ 3 = 1$
$(3 \circ 4) \circ 2 = (4 \circ 3) \circ 2 = 2 \circ (4 \circ 3) = 2 \circ (3 \circ 4) = 2 \circ 2 = 1$

All equal so assoc., thus $\boxed{\text{group}}$.

(c)

$\circ$	1	2	3	4
1	1	3	4	1
2	2	2	3	4
3	3	4	1	2
4	4	1	2	3

No identity, not a group.

(b)

$\circ$	1	2	3	4
1	1	2	3	4
2	2	3	4	1
3	3	4	1	2
4	4	1	2	3

$\circ$ is commutative

1 is identity
All elements have identity

$2 \circ (3 \circ 4) = 2 \circ 2 = 3$
$= 2 \circ (4 \circ 3)$
$= (4 \circ 3) \circ 2$
$= (3 \circ 4) \circ 2$

$(2 \circ 3) \circ 4 = 4 \circ 4 = 3$
$= (3 \circ 2) \circ 4$
$= 4 \circ (3 \circ 2)$
$= 4 \circ (2 \circ 3)$

$(2 \circ 4) \circ 3 = 1 \circ 3 = 3$
$= (4 \circ 2) \circ 3$
$= 3 \circ (4 \circ 2)$
$= 3 \circ (2 \circ 4)$

All equal, associative

$\boxed{\text{Group}}$

(d)

$\circ$	1	2	3	4
1	1	2	3	4
2	2	1	3	4
3	3	2	1	4
4	4	4	2	3

1 is identity
4 has no inverse

Not a group.

2. (12 points) Prove that $(S, \circ)$ is an abelian group where $S = \mathbb{R} \setminus \{-1\}$ and for any $x, y \in S$, $x \circ y = x + y + xy$.

Let $S = \mathbb{R} \setminus \{-1\}$ and define $\circ$ so that for any $x, y \in S$
$x \circ y = x + y + xy$.

Binary: The result $x + y + xy$ is always defined, and always a real number.

Notice $x + y + xy = -1$ only if $y(1+x) = -(1+x)$ $\boxed{\text{so } y = -1}$ but $y \in \mathbb{R} \setminus \{-1\}$

so $x + y + xy \in \mathbb{R} \setminus \{-1\}$.

$\circ$ is binary.

Identity: $x \circ e \overset{?}{=} x$ then $x + e + xe = x$

so $e + xe = 0$, $e(1+x) = 0$, $e = 0$

$0 \in S$, so the identity is 0.

Inverses: $x \circ x^{-1} \overset{?}{=} 0$, $x + x^{-1} + xx^{-1} = 0$

so $x^{-1}(1+x) = -x$ or $x^{-1} = -\dfrac{x}{1+x}$

since $x \neq -1$, x^{-1} is always defined and $x^{-1} \neq -1$ itself so inverses exist for all elements, $x^{-1} = -\dfrac{x}{1+x}$.

Assoc: $x \circ (y \circ z) \overset{?}{=} (x \circ y) \circ z$

$x \circ (y \circ z) = x \circ (y + z + yz) = x + y + z + yz + x(y + z + yz)$

$= x + y + z + yz + xy + xz + xyz = (x + y + xy) + z + (x + y + xy)z$

$= (x + y + xy) \circ z = (x \circ y) \circ z$. So $\circ$ is assoc.

Therefore $(S, \circ)$ is a group.

3. (12 points) Let $(G, \circ)$ be a group. Prove that the inverse of $g_1 \circ g_2 \circ \cdots \circ g_n$ is $g_n^{-1} \circ g_{n-1}^{-1} \circ \cdots \circ g_1^{-1}$ for any elements $g_1, g_2, ..., g_n$ in G.

Let $P(n)$= " $g_n^{-1} \circ g_{n-1}^{-1} \circ \cdots \circ g_1^{-1}$ is the inverse of $g_1 \circ g_2 \circ \cdots \circ g_n$ "

where $g_1, \dots, g_n$ are elements of a group $(G, \circ)$.

<u>Base Case</u> $(n=1)$ Consider $P(1)$: " g_1^{-1} is the inverse of g_1." This is true by definition so the base case holds.

<u>Inductive Step</u>: Let k be any positive integer and assume $P(k)$ is true. Then $(g_1 \circ \cdots \circ g_k)^{-1} = g_k^{-1} \circ \cdots \circ g_1^{-1}$.

We want to show $P(k+1)$ is also true, or that

$$(g_1 \circ \cdots \circ g_k \circ g_{k+1})^{-1} = g_{k+1}^{-1} \circ g_k^{-1} \circ \cdots \circ g_1^{-1}.$$

Consider the product

$$\left(g_1 \circ \cdots \circ g_k \circ g_{k+1}\right) \circ \left(g_{k+1}^{-1} \circ g_k^{-1} \circ \cdots \circ g_1^{-1}\right)$$

$$= \left(g_1 \circ \cdots \circ g_k\right) \circ \left(g_{k+1} \circ g_{k+1}^{-1}\right) \circ \left(g_k^{-1} \circ \cdots \circ g_1^{-1}\right)$$

$$= (g_1 \circ \cdots \circ g_k) \circ (g_k^{-1} \circ \cdots \circ g_1^{-1}) = e$$

by $P(k)$. Then $g_{k+1}^{-1} \circ g_k^{-1} \circ \cdots \circ g_1^{-1}$ is the inverse of $g_1 \circ g_2 \circ \cdots \circ g_k \circ g_{k+1}$.

Therefore, the inverse of $g_1 \circ \cdots \circ g_n$ is $g_n^{-1} \circ \cdots \circ g_1^{-1}$ for any elements $g_1, \dots, g_n$ in the group $(G, \circ)$ ∎

4. (12 points) Let S be the set defined as $S = \left\{ \begin{bmatrix} 1 & x & y \\ 0 & 1 & z \\ 0 & 0 & 1 \end{bmatrix} \middle|\, x, y, z \in \mathbb{R} \right\}$.

(a) Prove that $(S, \circ)$ is a group if $\circ$ is standard matrix multiplication.

(b) Is $(S, \circ)$ an abelian group? Justify your answer.